

**FORLENS – A LIGHTWEIGHT AI-DRIVEN ENDPOINT
SECURITY SOLUTION FOR REAL-TIME THREAT
DETECTION IN SMES**

25-26J-076

Individual Component Summary Report
(Lightweight Endpoint Agent Integrated with Honeypot Module)

Nusfa M R F – IT22908742

Bsc (Hons) in Information Technology Specializing in Cyber Security

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

**FORLENS – A LIGHTWEIGHT AI-DRIVEN ENDPOINT
SECURITY SOLUTION FOR REAL-TIME THREAT
DETECTION IN SMES**

25-26J-076

Individual Component Summary Report

(Lightweight Endpoint Agent Integrated with Honeypot Module)

Nusfa M R F – IT22908742

Bsc (Hons) in Information Technology Specializing in Cyber Security

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

Declaration of the Candidate & Supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Student Name	Student ID	Signature
NUSFA M.R.F	IT22908742	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Abstract

Small and Medium Enterprises (SMEs) are increasingly targeted by sophisticated cyber threats such as ransomware, lateral movement attacks, and unauthorized access attempts. However, due to financial and technical constraints, many SMEs are unable to deploy enterprise-grade security solutions such as Security Information and Event Management (SIEM) or Endpoint Detection and Response (EDR) systems. This creates a critical need for lightweight, cost-effective, and deployable endpoint-level security mechanisms. This report presents the design and implementation of a lightweight endpoint monitoring and containment agent developed as part of the ForLENS cybersecurity framework. The proposed agent integrates real-time process monitoring, file system monitoring, network connection tracking, and an embedded SMB-based honeypot mechanism to detect suspicious activities at the host level. Unlike traditional approaches, the deception mechanism is directly embedded within the endpoint, enabling early detection of lateral movement attempts without requiring additional infrastructure. The system incorporates signature-based detection using a predefined Indicator of Compromise (IOC) database, allowing immediate quarantine of malicious files. In addition, the agent supports offline operational capability by buffering logs locally during network disruptions and synchronizing them with a centralized backend upon reconnection. To enhance reliability, self-protection mechanisms such as watchdog-based process recovery and log regeneration are implemented to ensure continuous monitoring even under failure or tampering conditions. Experimental observations demonstrate that the proposed agent operates with low computational overhead while providing real-time detection and rapid containment capabilities. The results highlight the feasibility of integrating monitoring, deception, containment, and resilience within a single lightweight endpoint solution tailored for SME environments. Overall, this work demonstrates that practical and scalable cybersecurity protection can be achieved at the endpoint level without reliance on complex or resource-intensive infrastructure.

Keywords: *Lightweight Endpoint Agent, Integrated Honeypot, SME Cybersecurity, Threat Detection, Deception-Based Defense.*

Table of Content

Declaration of the Candidate & Supervisor	iii
Abstract	iv
Table of Content.....	v
List of Tables.....	vi
List of Figures	vi
List of Abbreviations	vi
1. INTRODUCTION	1
1.1 Background & Literature Survey.....	1
1.2 Research Gap	3
1.3 Research Problem	7
1.4 Research Objectives.....	9
2. METHODOLOGY	11
2.1 Methodology Overview	11
2.2 System Design and Architecture of Endpoint Agent	13
2.3 Testing and Implementation.....	15
3. METHODOLOGY	18
3.1 Overview of Endpoint Agent	18
3.2 Monitoring Modules	19
3.3 Multi-threaded Execution Design	22
3.4 Logging and Telemetry System	23
3.5 Signature-Based Detection and Quarantine Mechanism	25
3.6 Offline Logging and Synchronization.....	27
3.7 Self-Protection and Resilience Mechanisms.....	28
3.8 Secure Communication and Backend Integration.....	30
4.RESULTS AND DISCUSSIONS	32
4.1 Results (Demonstration Outcomes)	32
4.2 Research Findings	33
4.3 Discussion	34
5. CONCLUSION.....	36
Reference List	38

Appendices.....	41
Appendix A : Scope of The Study.....	41
Appendix B : Limitations.....	41
Appendix C : Risks and Mitgations	42
Appendix D : Ethical Considerations	42

List of Tables

Table 1 : Comparison of Previous Research and the proposed Lightweight Endpoint Agent	8
---	---

List of Figures

Figure 1: System Architecture Diagram	13
---	----

List of Abbreviations

Abbreviation	Description
SME	Small and Medium-Sized Enterprises
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation, and Response
EDR	Endpoint Detection and Response
AI	Artificial Intelligence
IDS	Intrusion Detection System
IOC	Indicator of Compromise
TLS	Transport Layer Security
SSL	Secure Sockets Layer
GDPR	General Data Protection Regulation
PCI-DSS	Payment Card Industry Data Security Standard
IDE	Integrated Development Environment
API	Application Programming Interface
VM	Virtual Machine (VirtualBox / VMware)

CPU	Central Processing Unit
RAM	Random Access Memory
SHA-256	Secure Hash Algorithm
JSON	JavaScript Object Notation

1. INTRODUCTION

1.1 Background & Literature Survey

Small and Medium Enterprises (SMEs) are increasingly becoming primary targets of modern cyberattacks due to their limited cybersecurity budgets, lack of skilled personnel, and absence of advanced security infrastructure. With the rapid growth of digital transformation and cloud adoption, SMEs are now exposed to a wide range of threats including ransomware, insider misuse, credential theft, lateral movement, and advanced persistent threats (APTs). Unlike large organizations that deploy enterprise-grade solutions such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Security Orchestration, Automation, and Response (SOAR), SMEs often rely on basic security tools that lack real-time visibility and automated response capabilities [20], [23].

Traditional security mechanisms are predominantly signature-based and reactive in nature, making them ineffective against zero-day attacks and evolving malware variants. As a result, there has been a growing shift toward behavior-based and anomaly detection techniques. Research in lightweight intrusion detection systems demonstrates that efficient threat detection can be achieved using resource-aware models that analyze system-level activities such as process execution patterns, file access behavior, and network communication [15], [16], [17]. These approaches aim to balance detection accuracy with minimal computational overhead, making them suitable for deployment in constrained environments.

In addition to behavioral monitoring, cyber deception techniques such as honeypots have emerged as powerful tools for identifying attacker behavior and collecting threat intelligence. Honeypots simulate vulnerable systems or services to attract attackers and observe their actions in a controlled environment. Studies have shown that integrating honeypots into security architectures can significantly improve detection of unknown threats and lateral movement attempts [4], [19]. High-interaction honeypots such as

HoneyWin further demonstrate the capability to capture detailed attacker behavior within enterprise environments [3], while adaptive honeypot frameworks dynamically adjust their behavior to improve detection effectiveness [6]. However, most existing honeypot deployments require dedicated infrastructure and are not designed for lightweight endpoint-level integration.

Another critical aspect of modern cybersecurity systems is the integrity and reliability of log data. Logs play a vital role in incident response, forensic analysis, and compliance. However, traditional logging mechanisms are often vulnerable to tampering, especially in scenarios involving insider threats or compromised systems. To address this, researchers have proposed tamper-evident logging mechanisms using cryptographic techniques such as Merkle tree chaining and secure append-only data structures [7], [9]. These approaches ensure that any modification to log data can be detected, thereby preserving the integrity and trustworthiness of recorded events. Additionally, scalable logging architectures have been explored to support large-scale data collection and verification in distributed systems [8].

Furthermore, the increasing complexity of cyber threats has led to the development of automated incident response mechanisms aimed at reducing response time and minimizing manual intervention. Modern approaches in Security Orchestration, Automation, and Response (SOAR) focus on integrating detection with automated mitigation actions such as isolating compromised systems, blocking malicious processes, and generating alerts for analysts [10], [11]. While these systems are effective in enterprise environments, their deployment remains challenging for SMEs due to high cost, complexity, and infrastructure requirements.

Despite the significant advancements in behavioral detection, cyber deception, secure logging, and automated response, existing solutions often operate as isolated components rather than integrated systems. There is a lack of lightweight, unified solutions that combine real-time monitoring, embedded deception, local detection, and automated

containment within a single endpoint-level architecture. This limitation is particularly critical for SMEs, where resource constraints demand efficient, scalable, and easy-to-deploy security mechanisms.

To address these challenges, this work focuses on the development of a lightweight endpoint agent that integrates continuous system monitoring, deception-based detection through an embedded honeypot, signature-based containment, and resilient logging mechanisms. By consolidating these capabilities into a single deployable component, the proposed approach aims to provide practical and effective endpoint security tailored specifically for SME environments.

1.2 Research Gap

Despite the growing body of research in cybersecurity, there remains a significant gap in the development of lightweight, endpoint-level security solutions that can provide comprehensive monitoring, detection, and response capabilities within a single deployable framework. Existing approaches in intrusion detection systems (IDS) and endpoint monitoring primarily focus on isolated aspects of system behavior, such as network traffic analysis or anomaly detection, without offering a unified mechanism that captures process activity, file system changes, and network interactions simultaneously. Lightweight intrusion detection mechanisms proposed in recent studies aim to reduce computational overhead [15], [16], [17], yet they often lack integration across multiple monitoring domains, limiting their effectiveness in detecting complex, multi-stage attacks at the endpoint level.

In parallel, cyber deception techniques such as honeypots have demonstrated strong potential in identifying attacker behavior and detecting unauthorized access attempts [4], [19]. Advanced implementations, including high-interaction honeypots and adaptive deception frameworks, provide detailed insights into attacker tactics and techniques [3], [6], [18]. However, these systems are predominantly deployed as standalone infrastructure components or within virtualized environments, requiring dedicated resources and

complex configurations [2], [12]. This architectural design creates a barrier for practical deployment in SME environments and fails to address threats that originate directly at the host level, such as insider attacks and lateral movement. The integration of deception mechanisms directly into lightweight endpoint agents remains insufficiently explored, representing a critical gap in current research.

Another key limitation lies in the lack of immediate containment capabilities within endpoint-level monitoring systems. While detection techniques have evolved significantly, many existing solutions rely on centralized platforms or external orchestration systems to initiate response actions [10], [11]. This introduces latency between detection and mitigation, which can be critical in fast-moving threats such as ransomware or malicious file execution. The absence of integrated, local containment mechanisms within lightweight endpoint agents highlights the need for systems that can perform both detection and response directly at the host level without dependency on centralized infrastructure.

Furthermore, resilience and reliability remain under-addressed in current endpoint monitoring solutions. Many systems assume continuous network connectivity for data transmission and analysis, which is not always feasible in real-world SME environments. Network disruptions can lead to data loss, incomplete logging, and gaps in monitoring coverage. Although offline buffering and synchronization mechanisms are essential for maintaining operational continuity, they are not commonly integrated into lightweight endpoint solutions. This creates a vulnerability in maintaining consistent security visibility under unstable network conditions.

In addition to operational limitations, logging mechanisms in many existing systems lack robustness against tampering and accidental deletion. Logs play a crucial role in forensic analysis and incident response, yet traditional logging approaches are susceptible to modification when an attacker gains system-level access. Research in tamper-evident logging using cryptographic structures such as Merkle trees demonstrates the importance of maintaining data integrity [7], [9], but such mechanisms are often implemented in large-

scale or centralized systems rather than lightweight endpoint environments. The lack of resilient, self-maintaining logging systems at the endpoint level further contributes to the existing research gap.

Overall, current cybersecurity solutions tend to address monitoring, deception, detection, logging, and response as separate components rather than integrating them into a unified, lightweight endpoint architecture. This fragmentation increases system complexity, limits deployment feasibility, and reduces effectiveness in resource-constrained environments. Therefore, there is a clear need for a lightweight endpoint agent that combines real-time multi-layer monitoring, embedded deception mechanisms, local detection and containment, and resilient logging within a single system. Addressing this gap is essential for developing practical and deployable cybersecurity solutions tailored to the operational constraints and security requirements of SMEs.

Features/Contributions	Research [3]	Research [2],[6]	Research [15],[16]	Research [5]	Research [4],[13]	Proposed System
Low-interaction honeypot integrated into endpoint agent	X	X	X	X	X	✓
Lightweight deception-based honeypot	X	✓	X	X	✓	✓
Generates context rich telemetry for integration with anomaly detection	X	X	✓	X	✓	✓

Stealth mode / tamper-resistant operation	X	X	X	✓	X	✓
Real-time event forwarding	X	X	X	X	X	✓
Supports correlation by forwarding honeypot + endpoint telemetry	X	X	X	X	X	✓
Plug-and-play deployment (low technical effort)	X	X	X	X	X	✓
Tailored for SME specific attack surfaces	X	X	X	X	X	✓
Cost-effective, easy to-deploy solution for SMEs	X	X	X	X	X	✓
Local signature- based detection	X	X	X	X	X	✓
Automated containment (quarantine)	X	X	X	X	X	✓

Watchdog-based self-recovery	X	X	X	X	X	✓
Offline logging and synchronization	X	X	X	X	X	✓

Table 1 : Comparison of Previous Research and the proposed Lightweight Endpoint Agent

1.3 Research Problem

Despite the availability of advanced cybersecurity solutions, Small and Medium Enterprises (SMEs) continue to face significant challenges in deploying effective endpoint-level security mechanisms. Most existing solutions such as enterprise-grade endpoint protection platforms and intrusion detection systems are designed for large-scale environments and rely on centralized architectures, making them costly and complex for SMEs to adopt [20], [23]. This creates a practical limitation where SMEs lack continuous visibility into endpoint activities, leaving them vulnerable to modern cyber threats.

A key research problem lies in the absence of unified endpoint-level monitoring systems capable of simultaneously capturing process activity, file system changes, and network interactions. Existing lightweight intrusion detection mechanisms focus on specific domains such as network-based or anomaly-based detection, without providing a comprehensive view of endpoint behavior [15], [16], [17]. This limitation reduces their effectiveness in detecting multi-stage attacks that involve coordinated activities across different system layers.

In addition, current research does not adequately address the integration of deception mechanisms within endpoint environments. While honeypots have been widely studied and proven effective in detecting attacker behavior and capturing threat intelligence [4], [19], most implementations are deployed as standalone systems or within network infrastructures [2], [12]. Advanced honeypot models such as high-interaction and adaptive

honeypots further demonstrate the benefits of deception-based detection [3], [6], [18], yet their deployment remains resource-intensive and unsuitable for lightweight endpoint integration. As a result, there is a lack of practical solutions that embed deception directly within endpoint agents to detect host-level threats such as lateral movement and unauthorized access.

Another critical issue is the lack of immediate response capabilities at the endpoint level. Although automated incident response and orchestration mechanisms have been proposed to reduce response time [10], [11], these approaches are primarily designed for centralized security operations. This introduces delays between detection and mitigation, which can be critical in fast-moving attacks such as ransomware. The absence of local containment mechanisms within lightweight endpoint systems highlights the need for solutions that can perform real-time detection and immediate response directly at the host level.

Furthermore, many existing endpoint monitoring solutions depend heavily on continuous network connectivity for log transmission and analysis. In real-world SME environments, network instability can lead to interruptions in monitoring and loss of critical data. Current systems often lack offline operational capability, resulting in incomplete visibility during connectivity disruptions. This emphasizes the need for endpoint agents that can operate independently, store logs locally, and synchronize data once connectivity is restored.

Another important problem relates to the reliability and integrity of logging mechanisms. Logs are essential for forensic analysis and incident response; however, traditional logging systems are vulnerable to tampering and deletion when attackers gain access to the system. Research in tamper-evident logging techniques, such as Merkle tree-based approaches, highlights the importance of preserving log integrity [7], [9]. However, these mechanisms are rarely integrated into lightweight endpoint-level solutions, creating a gap in ensuring reliable and continuous logging in constrained environments.

Overall, the core research problem addressed in this work is the lack of a lightweight, integrated, and resilient endpoint-level security solution that combines multi-layer monitoring, deception-based detection, real-time containment, and reliable logging within

a single deployable agent. Addressing this problem is essential to provide SMEs with an effective and practical cybersecurity solution that does not rely on complex or resource-intensive infrastructure.

1.4 Research Objectives

The primary objective of this work is to design and implement a lightweight endpoint-level monitoring and containment agent that provides real-time visibility, detection, and response capabilities within resource-constrained environments. The system aims to operate efficiently at the host level while minimizing dependency on centralized infrastructure.

To achieve this, the following specific objectives are defined:

- To develop a lightweight endpoint agent capable of continuously monitoring process activity, file system changes, and network connections in real time.
- To integrate a deception-based detection mechanism using an embedded low-interaction SMB honeypot to identify unauthorized access attempts and lateral movement behavior.
- To implement a signature-based detection mechanism using a local Indicator of Compromise (IOC) database for identifying known malicious files.
- To design and implement an automated containment mechanism that performs immediate quarantine of detected threats at the endpoint level.
- To develop a structured logging and telemetry system that captures context-rich event data for analysis and correlation.
- To ensure system reliability through offline logging and synchronization, enabling continuous monitoring during network disruptions.
- To incorporate self-protection and resilience mechanisms, including watchdog-based process recovery and log regeneration, to maintain uninterrupted operation.
- To enable secure communication with a backend system using encrypted channels for centralized analysis and data aggregation.

- To ensure that the overall system operates with low computational overhead, making it suitable for deployment in SME environments.

2. METHODOLOGY

2.1 Methodology Overview

The proposed component focuses on the design and implementation of a lightweight endpoint monitoring and containment agent capable of providing real-time visibility, detection, and response within resource-constrained environments. The methodology adopts a modular and efficient approach to ensure continuous monitoring of endpoint activities while maintaining low computational overhead, making it suitable for deployment in Small and Medium Enterprise (SME) environments.

The core design principle of the system is based on lightweight telemetry collection combined with localized detection and response mechanisms. Unlike traditional approaches that rely heavily on centralized infrastructures, the proposed methodology enables monitoring, detection, and containment directly at the endpoint level. This approach reduces dependency on external systems and improves response time, particularly in environments with limited connectivity or infrastructure support.

To achieve comprehensive endpoint visibility, the system employs multi-layer monitoring by capturing process execution events, file system activities, network connections, and unauthorized access attempts through an embedded deception mechanism. This multi-dimensional monitoring strategy ensures that different stages of an attack lifecycle can be observed and analyzed in real time. The design aligns with research on lightweight intrusion detection systems that emphasize efficient monitoring with minimal resource utilization.

A key aspect of the methodology is the integration of a deception-based detection mechanism through an embedded SMB honeypot within the endpoint agent. Unlike traditional honeypot deployments that require separate infrastructure, this approach embeds the deception layer directly within the host environment, enabling detection of lateral movement and unauthorized access attempts at the source. This design is inspired by cyber deception techniques that enhance threat detection by attracting and monitoring attacker interactions.

In addition to monitoring and deception, the methodology incorporates a signature-based detection mechanism using a predefined Indicator of Compromise (IOC) database. The system computes cryptographic hash values of files and compares them against known malicious signatures to identify threats. Upon detection, an automated containment process is triggered, enabling immediate quarantine of suspicious files at the endpoint. This approach aligns with automated incident response concepts aimed at reducing detection-to-response latency.

To ensure reliability and continuity of operations, the methodology also integrates resilience mechanisms such as offline log buffering and watchdog-based self-recovery. The system is designed to operate independently during network disruptions by storing events locally and synchronizing them with a centralized backend upon reconnection. Additionally, the watchdog mechanism ensures that the monitoring agent is automatically restarted in the event of unexpected termination, enhancing system survivability.

Furthermore, the system incorporates secure communication protocols to transmit structured telemetry data to the backend analysis engine. Data is transmitted over encrypted HTTPS channels, ensuring confidentiality and integrity during transmission. This enables seamless integration with centralized systems for further analysis and correlation while maintaining secure data exchange.

Overall, the methodology combines lightweight monitoring, embedded deception, local detection and containment, resilient logging, and secure communication within a unified endpoint agent. This integrated approach addresses the limitations of existing fragmented solutions and provides a practical, efficient, and deployable cybersecurity mechanism tailored for SME environments.

2.2 System Design and Architecture of Endpoint Agent

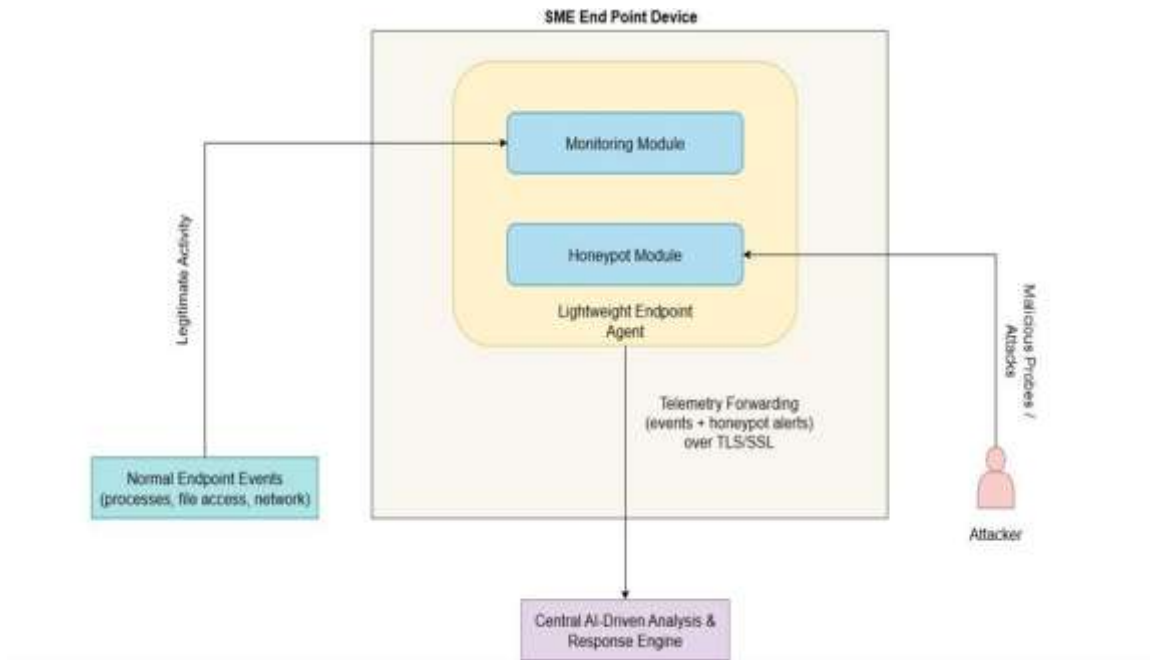


Figure 1: System Architecture Diagram

The proposed endpoint agent is designed as a lightweight, modular, and resilient system that operates directly at the host level to provide continuous monitoring, detection, and response capabilities. The architecture follows a component-based design, where each functional module is independently implemented but integrated through a unified event processing and logging pipeline. This modular approach ensures scalability, maintainability, and efficient execution within resource-constrained environments.

At a high level, the system architecture consists of multiple interconnected layers, including telemetry collection, deception, detection and containment, logging, communication, and resilience. These layers collectively enable the agent to capture system activities, identify malicious behavior, respond to threats, and maintain operational continuity.

- The **Telemetry Collection Layer** is responsible for continuously monitoring endpoint activities. It captures system-level events such as process creation and termination, file system modifications, network connections, and share access

attempts. These events provide the foundational data required for detecting suspicious behavior. The monitoring approach is designed to be lightweight, avoiding computationally intensive operations while still ensuring sufficient visibility into system activities. This aligns with lightweight monitoring principles explored in intrusion detection research

- The **Deception Layer** embeds a low-interaction SMB-based honeypot directly within the endpoint environment. This module acts as a decoy by simulating a vulnerable network share, attracting unauthorized access attempts and lateral movement behavior. When an attacker interacts with the decoy resource, the system captures critical information such as source IP address, access time, and system context. Unlike traditional honeypot architectures that require dedicated infrastructure [3], this embedded approach reduces deployment complexity while maintaining effective detection capability. The use of deception mechanisms enhances attacker visibility and aligns with research in cyber deception strategies.
- The **Detection and Containment Layer** performs analysis on the collected telemetry using signature-based and rule-based techniques. The system computes SHA-256 hash values of files and compares them against a predefined IOC database to identify known malicious artifacts. Upon detection, the containment module initiates automated mitigation actions such as isolating and quarantining suspicious files. This localized response mechanism eliminates dependency on centralized systems and reduces response time, reflecting principles from automated incident response frameworks.
- The **Logging Layer** ensures that all events are recorded in a structured and reliable manner. Logs are stored locally in a JSON-based format, allowing easy integration with backend systems. The logging mechanism is designed to maintain data integrity and continuity, even in the presence of unexpected failures or deletion attempts. Although full cryptographic chaining mechanisms such as Merkle trees are not implemented due to performance considerations, the design incorporates integrity-aware principles inspired by tamper-evident logging research.

- The **Communication Layer** is responsible for transmitting collected telemetry data to a centralized analysis engine. Data is sent using HTTPS-based secure communication channels, ensuring confidentiality and integrity during transmission. In scenarios where network connectivity is unavailable, the system temporarily stores logs locally and automatically synchronizes them once the connection is restored. This offline capability ensures uninterrupted monitoring and prevents data loss, which is particularly important in SME environments with unstable connectivity.
- The **Resilience and Self-Recovery Layer** enhances the robustness of the system by ensuring continuous operation. A watchdog mechanism continuously monitors the status of the agent and automatically restarts it in case of unexpected termination. Additionally, the system includes log regeneration mechanisms to recover from accidental deletion of log files. These features ensure that monitoring and logging operations remain active even under adverse conditions, improving overall system survivability.

The interaction between these layers forms a unified architecture where telemetry flows from the monitoring modules to the detection engine, triggering containment actions when necessary, while simultaneously recording events and transmitting them securely to the backend. By integrating monitoring, deception, detection, containment, logging, and resilience within a single endpoint-level framework, the proposed architecture provides a comprehensive yet lightweight solution tailored for SME cybersecurity requirements.

2.3 Testing and Implementation

The proposed endpoint agent was implemented as a modular, lightweight system designed to operate efficiently within a Windows-based endpoint environment. The implementation focused on achieving real-time monitoring, detection, and containment while maintaining low resource consumption and minimal system interference. A Python-based development approach was adopted due to its flexibility, availability of system-level libraries, and suitability for rapid prototyping of security monitoring tools.

The system was developed using a combination of standard libraries and lightweight modules, including *psutil* for process and network monitoring, *os* and *hashlib* for file system tracking and hash generation, *threading* for concurrent execution of monitoring components, and *requests* for secure communication with the backend system. These technologies enabled the development of a multi-threaded architecture in which each monitoring module operates independently while contributing to a unified event collection pipeline.

To ensure comprehensive monitoring, the implementation includes multiple modules responsible for capturing different types of system activity. The process monitoring module continuously tracks process creation and termination events, collecting metadata such as process identifiers, executable paths, and command-line arguments. The file monitoring module observes file system changes, including creation, modification, and deletion events, and computes SHA-256 hash values for integrity verification and threat detection. The network monitoring module captures active network connections and associated process information, enabling visibility into communication patterns.

A key feature of the implementation is the embedded SMB-based deception mechanism. A decoy network share is configured within the endpoint environment to simulate a vulnerable resource. Unauthorized access attempts to this decoy are detected by monitoring relevant system events, allowing the system to identify potential lateral movement or reconnaissance activity. This approach eliminates the need for external honeypot infrastructure while maintaining effective detection capability.

The detection mechanism is implemented using a signature-based approach, where computed file hashes are compared against a predefined Indicator of Compromise (IOC) database. When a match is identified, the system triggers an automated containment process. The containment mechanism isolates the suspicious file by moving it to a secure quarantine directory and logs the action for further analysis. This local containment strategy ensures rapid response without reliance on centralized orchestration systems, aligning with automated incident response principles [10], [11].

Testing of the system was conducted using controlled simulation scenarios to validate detection accuracy, response time, and system reliability. These scenarios included unauthorized access to the honeypot resource, execution of files with known malicious signatures, and manual attempts to disrupt system operations such as deleting log files or terminating the agent process. The system successfully detected suspicious activities, triggered containment actions, and maintained operational continuity under these test conditions.

To evaluate resilience, the agent was tested under network disruption conditions by disabling connectivity between the endpoint and the backend system. During this period, events were stored locally and successfully transmitted once connectivity was restored, demonstrating reliable offline operational capability. Additionally, the watchdog mechanism was tested by forcibly terminating the agent process, after which the system automatically restarted and resumed monitoring without manual intervention.

Performance considerations were incorporated throughout the implementation process to ensure suitability for SME environments. The system avoids computationally intensive operations such as continuous deep packet inspection or heavy machine learning inference, instead relying on efficient event monitoring and rule-based detection. As a result, the agent operates with low CPU and memory usage, ensuring minimal impact on normal system performance.

Overall, the implementation and testing demonstrate that the proposed endpoint agent effectively integrates monitoring, deception, detection, containment, and resilience within a single lightweight system. The results confirm that the system can operate reliably in real-world conditions while maintaining efficiency and scalability for SME deployment

3. METHODOLOGY

3.1 Overview of Endpoint Agent

The design of the suggested solution focuses on a Lightweight Endpoint Agent installed on SME endpoints. The agent consists of two main components:

The proposed endpoint agent is designed as a lightweight, host-based security component that provides continuous visibility, detection, and response capabilities directly at the endpoint level. Unlike traditional security solutions that rely on centralized architectures, the agent operates independently on each endpoint, enabling real-time monitoring and immediate threat handling without dependency on external systems.

The primary objective of the endpoint agent is to collect system-level telemetry, identify suspicious activities, and initiate local containment actions while maintaining minimal impact on system performance. To achieve this, the agent integrates multiple functional capabilities, including process monitoring, file system tracking, network activity observation, and deception-based detection through an embedded honeypot mechanism.

The agent is implemented as a unified system where all monitoring components operate concurrently, enabling continuous observation of endpoint behavior. Each module contributes to a centralized logging and event processing pipeline, ensuring that all captured activities are structured, recorded, and made available for further analysis. This integrated design eliminates the need for separate tools or infrastructure, making the system suitable for deployment in resource-constrained environments such as SMEs.

A key feature of the endpoint agent is its ability to perform detection and response locally. The system utilizes a signature-based detection approach by comparing file hash values against a predefined Indicator of Compromise (IOC) database. When a match is identified, the agent immediately triggers a containment action by isolating the suspicious file. This localized response capability significantly reduces the time between detection and mitigation, which is critical in preventing the spread of fast-acting threats.

In addition to detection, the agent incorporates a deception-based mechanism through an embedded SMB honeypot. This allows the system to detect unauthorized access attempts and lateral movement activities by monitoring interactions with decoy resources. By integrating deception directly within the endpoint, the system enhances its ability to detect threats that may bypass traditional monitoring techniques.

The agent also includes resilience and reliability features to ensure continuous operation. It supports offline logging by storing events locally during network disruptions and synchronizing them with the backend system once connectivity is restored. Furthermore, a watchdog mechanism is implemented to automatically restart the agent in case of unexpected termination, ensuring uninterrupted monitoring.

Overall, the endpoint agent functions as a comprehensive, lightweight security component that combines monitoring, detection, deception, containment, and resilience within a single deployable unit. This design makes it particularly suitable for SME environments, where simplicity, efficiency, and cost-effectiveness are critical requirements.

3.2 Monitoring Modules

The endpoint agent incorporates multiple monitoring modules designed to capture different types of system-level activities in real time. These modules operate concurrently

•

and collectively provide comprehensive visibility into endpoint behavior. Each module focuses on a specific domain of system activity, including process execution, file system changes, network communication, and unauthorized access attempts through a deception mechanism. The modular design ensures scalability and allows each monitoring component to function independently while contributing to a unified telemetry and logging pipeline.

3.2.1 Process Monitoring Module

The process monitoring module is responsible for continuously tracking process-related activities within the system. It monitors events such as process creation and termination and captures associated metadata including process identifiers (PID), parent process identifiers (PPID), executable paths, command-line arguments, and user context. This information is essential for identifying suspicious behavior patterns such as unauthorized process execution, abnormal parent-child relationships, and privilege misuse.

The module utilizes lightweight system-level libraries to retrieve process information in real time, ensuring minimal impact on system performance. By maintaining continuous visibility into process activity, the system can detect anomalies such as unexpected process launches or execution from unusual directories. All captured events are structured and forwarded to the logging system for further analysis and correlation.

3.2.2 File Monitoring Module

The file monitoring module is designed to observe file system activities, including file creation, modification, and deletion events. It continuously monitors critical directories and tracks changes to files that may indicate suspicious behavior, such as unauthorized file drops or modifications commonly associated with malware execution.

For each observed file event, the module computes a SHA-256 hash value to uniquely identify the file and support integrity verification. These hash values are later used in the

detection process by comparing them against a predefined Indicator of Compromise (IOC) database. This enables the system to identify known malicious files efficiently.

The module is optimized to avoid excessive resource usage by focusing on relevant directories and performing event-driven monitoring rather than continuous full-system scanning. All file-related events and associated metadata are recorded and passed to the logging and detection components.

3.2.3 Network Monitoring Module

The network monitoring module provides visibility into active network connections and communication patterns within the system. It captures details such as local and remote IP addresses, ports, connection states, and associated process identifiers. This allows the system to correlate network activity with specific processes, enabling detection of suspicious outbound connections or unauthorized communication attempts.

By monitoring network connections in real time, the module helps identify potential command-and-control (C2) communication, data exfiltration attempts, or connections to known malicious endpoints. The collected network data is integrated into the overall telemetry pipeline, allowing correlation with process and file activity for enhanced threat analysis.

The module is implemented using lightweight techniques to ensure efficient monitoring without introducing significant performance overhead, making it suitable for continuous operation in SME environments.

3.2.4 SMB Honeypot Module

The SMB honeypot module introduces a deception-based detection mechanism directly within the endpoint environment. It creates a decoy resource that simulates a vulnerable network share, designed to attract unauthorized access attempts and lateral movement activities. When an attacker or malicious process interacts with this decoy, the system captures relevant details such as source IP address, access time, and system context.

•

Unlike traditional honeypot systems that require separate infrastructure, this module is embedded within the endpoint agent, enabling detection at the host level. This significantly improves the ability to detect insider threats and lateral movement attempts that may not be visible through conventional monitoring methods.

The honeypot operates as a low-interaction deception mechanism, ensuring minimal resource consumption while still providing valuable threat intelligence. All detected interactions are logged and forwarded for further analysis, contributing to improved situational awareness and early threat detection

3.3 Multi-threaded Execution Design

The endpoint agent is designed using a multi-threaded architecture to enable concurrent execution of all monitoring modules without blocking system performance. Since the agent is required to continuously monitor multiple system activities such as processes, files, network connections, and honeypot interactions, a sequential execution model would introduce delays and reduce the effectiveness of real-time monitoring. To address this, each monitoring component is executed as an independent thread, allowing simultaneous data collection across multiple domains.

The main execution flow of the agent initializes separate threads for each monitoring module, including the process monitoring module, file monitoring module, network monitoring module, and the SMB honeypot module. Once initialized, all threads are started together and run in parallel in the background. This design ensures that monitoring operations occur continuously and independently, without waiting for other modules to complete execution.

The use of multi-threading enables efficient utilization of system resources by distributing tasks across concurrent execution paths. Each thread is responsible for a specific monitoring function and operates within its own execution loop, capturing events in real

time and forwarding them to the logging system. This approach minimizes latency in data collection and ensures that critical events are not missed due to blocking operations in other modules.

Additionally, the threads are configured to run as daemon processes, allowing them to operate in the background without requiring direct user interaction. This supports the stealth-oriented design of the agent, where monitoring activities are performed silently without interrupting normal system usage. The main program maintains a persistent execution loop to keep all monitoring threads active throughout the system runtime.

The multi-threaded architecture also enhances the scalability and extensibility of the system. New monitoring modules or additional functionalities can be integrated as separate threads without affecting the existing system structure. This modular execution model allows the agent to evolve while maintaining performance efficiency.

Overall, the multi-threaded design plays a critical role in enabling real-time, continuous, and efficient monitoring across multiple system components. By allowing parallel execution of monitoring tasks, the agent achieves high responsiveness and ensures comprehensive visibility into endpoint activities while maintaining low performance overhead.

3.4 Logging and Telemetry System

The logging and telemetry system plays a critical role in the functionality of the endpoint agent by ensuring that all monitored activities are captured, structured, and made available for analysis. The system is designed to generate context-rich telemetry by collecting detailed information from multiple monitoring modules, including process activity, file system changes, network connections, and honeypot interactions. This unified logging approach enables comprehensive visibility into endpoint behavior and supports effective threat detection and analysis.

•

All events captured by the monitoring modules are processed through a centralized logging mechanism that formats the data into structured records. The logs are stored locally in a JSON-based format, allowing each event to be recorded as an individual entry with clearly defined attributes such as timestamp, event type, process information, file details, network data, and user context. This structured approach ensures consistency and simplifies integration with backend systems for further analysis and correlation.

The telemetry generated by the system is designed to be context-aware, meaning that each event includes sufficient metadata to support deeper analysis. For example, process-related events include details such as process identifiers, execution paths, and command-line arguments, while network events include source and destination information along with associated processes. This level of detail enables correlation across multiple data sources, which is essential for identifying complex attack patterns and multi-stage threats.

In addition to local logging, the system supports real-time telemetry transmission to a centralized backend analysis engine. Events are forwarded using HTTP-based requests over secure communication channels, enabling near real-time visibility beyond the endpoint. This allows the backend system to perform advanced analysis, correlation, and long-term storage of security events. The design aligns with modern security architectures that emphasize centralized analysis of distributed telemetry for improved threat intelligence

To ensure reliability, the logging system is designed with resilience in mind. Logs are continuously written to a local file, ensuring that no data is lost during normal operation. In scenarios where the log file is accidentally deleted or corrupted, the system includes mechanisms to automatically recreate the log file and continue recording events without interruption. This ensures continuity of monitoring and prevents loss of critical forensic data.

Although full cryptographic enforcement is not implemented due to performance considerations, the design follows principles of secure and integrity-aware logging inspired by tamper-evident logging research. By maintaining structured, consistent, and continuously updated logs, the system enhances trustworthiness and supports reliable incident investigation.

Overall, the logging and telemetry system serves as the backbone of the endpoint agent by enabling structured data collection, real-time event forwarding, and reliable storage of security events. This integrated approach ensures that all monitoring activities are effectively captured and made available for detection, correlation, and response processes within the system.

3.5 Signature-Based Detection and Quarantine Mechanism

The endpoint agent incorporates a signature-based detection mechanism to identify known malicious files and initiate immediate containment actions at the host level. This approach relies on the use of a predefined Indicator of Compromise (IOC) database, which stores cryptographic hash values of known malicious artifacts. By leveraging hash-based detection, the system is able to efficiently and accurately identify threats with minimal computational overhead.

When a file-related event is captured by the monitoring modules, particularly during file creation or execution, the system computes a SHA-256 hash of the file using a lightweight cryptographic function. This hash uniquely represents the file's content and is compared against the entries stored in the IOC database. If a match is identified, the file is classified as malicious, and the detection event is immediately triggered. This process enables rapid identification of known threats without requiring complex behavioral analysis.

Following detection, the system initiates an automated containment process designed to prevent further execution or propagation of the malicious file. The containment mechanism isolates the identified file by moving it to a designated quarantine directory

•

within the system. This ensures that the file is no longer accessible for execution while preserving it for further analysis if required. The quarantine action is performed locally at the endpoint, eliminating delays associated with centralized response systems and significantly reducing the time between detection and mitigation.

All detection and containment events are recorded within the logging system, including details such as file path, computed hash value, detection timestamp, and action taken. These logs provide a clear audit trail for forensic analysis and enable further investigation into the source and impact of the detected threat. Additionally, the events can be forwarded to the backend system for centralized monitoring and correlation with other telemetry data.

The implementation of local signature-based detection combined with automated quarantine aligns with principles of rapid incident response, where minimizing response time is critical to limiting damage caused by threats such as ransomware or unauthorized file execution. By performing both detection and containment directly at the endpoint, the system enhances its ability to respond effectively in real time without reliance on external infrastructure.

Furthermore, the design ensures that the detection mechanism remains lightweight and efficient. Unlike advanced machine learning-based detection systems, which may introduce computational overhead, the hash-based approach provides a practical balance between accuracy and performance. This makes it particularly suitable for deployment in SME environments where resource efficiency is a key requirement.

Overall, the signature-based detection and quarantine mechanism enables the endpoint agent to provide fast, reliable, and localized threat mitigation. By integrating detection and containment within a single workflow, the system ensures immediate response to known threats while maintaining simplicity and efficiency in operation.

3.6 Offline Logging and Synchronization

The endpoint agent is designed to operate reliably even in environments where continuous network connectivity cannot be guaranteed. To address this, the system incorporates an offline logging and synchronization mechanism that ensures uninterrupted monitoring and prevents loss of critical telemetry data during network disruptions. This capability is particularly important for Small and Medium Enterprise (SME) environments, where network instability or limited infrastructure is a common challenge [20].

During normal operation, the agent continuously transmits collected telemetry data to a centralized backend system using secure communication channels. However, when the connection to the backend becomes unavailable, the system automatically switches to offline mode without interrupting monitoring activities. In this state, all events generated by the monitoring modules are stored locally within the endpoint in a structured log format.

The offline logging mechanism ensures that all captured events, including process activity, file system changes, network connections, and honeypot interactions, are recorded without any data loss. Each event is appended to a local log file, preserving the sequence and integrity of the collected telemetry. This design guarantees continuous visibility into endpoint activity, even when external communication is temporarily disrupted.

Once network connectivity is restored, the system initiates a synchronization process in which all locally stored events are transmitted to the backend system. The synchronization is performed automatically without requiring user intervention, ensuring seamless recovery of telemetry data. This process maintains consistency between local and centralized records and enables the backend system to perform comprehensive analysis on previously buffered events.

•

The implementation of offline buffering and delayed transmission aligns with the need for resilient telemetry systems in distributed environments. By decoupling monitoring from communication, the agent ensures that security operations are not dependent on real-time connectivity. This design improves reliability and enhances the robustness of the overall system.

In addition to ensuring data continuity, the offline synchronization mechanism also supports efficient resource usage. Instead of repeatedly attempting to send data during connectivity failures, the system stores events locally and transmits them in batches once connectivity is restored. This reduces unnecessary network overhead and improves overall system efficiency.

Overall, the offline logging and synchronization feature enhances the practicality and reliability of the endpoint agent by ensuring that monitoring remains active under all conditions. It guarantees that no critical security events are lost and that full telemetry is available for analysis once communication with the backend system is re-established.

3.7 Self-Protection and Resilience Mechanisms

To ensure continuous operation and resistance against interruption or tampering, the endpoint agent incorporates self-protection and resilience mechanisms designed to maintain monitoring integrity under adverse conditions. In real-world attack scenarios, adversaries often attempt to disable security tools, terminate monitoring processes, or manipulate system logs to evade detection. Therefore, the proposed system integrates defensive mechanisms that enhance survivability and ensure uninterrupted functionality.

A key component of the resilience design is the implementation of a watchdog-based self-recovery mechanism. The watchdog operates as a dedicated monitoring process that continuously checks the operational status of the main endpoint agent. If the agent is unexpectedly terminated due to system failure or malicious activity, the watchdog automatically restarts the agent without requiring manual intervention. This ensures that

monitoring is quickly restored and minimizes downtime. The inclusion of automated recovery aligns with resilience-focused security principles that emphasize maintaining system availability even under attack conditions [5], [8].

In addition to process recovery, the system incorporates log continuity protection to prevent loss of critical monitoring data. Since logs are essential for forensic analysis and incident investigation, the agent is designed to handle scenarios where the local log file may be deleted or corrupted. If such an event occurs, the system automatically regenerates the log file and continues recording events without interruption. This mechanism ensures that logging operations remain consistent and that monitoring data is preserved despite accidental deletion or intentional tampering attempts.

The agent also supports persistent background execution, enabling it to operate continuously without requiring user interaction. Monitoring modules run silently in the background as daemon threads, contributing to the stealth-oriented design of the system. This reduces the likelihood of accidental disruption and ensures that security monitoring does not interfere with normal user activities.

Furthermore, the resilience design ensures compatibility with system startup mechanisms, allowing the agent to be configured for automatic execution during system boot. This guarantees that monitoring begins immediately when the system is active, providing continuous protection from the initial stage of operation.

The combination of watchdog-based recovery, log regeneration, and persistent background execution creates a robust self-protection framework for the endpoint agent. These mechanisms ensure that the system remains operational, reliable, and resistant to disruption, even in the presence of malicious interference or system instability.

Overall, the integration of self-protection and resilience features enhances the robustness of the endpoint agent by ensuring continuous monitoring, maintaining data integrity, and minimizing operational downtime. This makes the system well-suited for real-world

•

deployment in SME environments, where reliability and simplicity are critical requirements.

3.8 Secure Communication and Backend Integration

The endpoint agent is designed to integrate seamlessly with a centralized backend system for advanced analysis, correlation, and long-term storage of security events. To achieve this, the system incorporates a secure communication mechanism that enables reliable transmission of telemetry data from the endpoint to the backend while ensuring data confidentiality and integrity.

All telemetry generated by the monitoring modules is structured and transmitted using HTTP-based RESTful communication over secure HTTPS channels. The use of HTTPS ensures that all data exchanged between the endpoint and the backend is encrypted using Transport Layer Security (TLS), protecting it from interception, modification, or unauthorized access during transmission. This is particularly important in distributed environments where telemetry data may traverse untrusted networks.

The communication module is responsible for formatting collected events into structured JSON objects and sending them to the backend system through defined API endpoints. Each event includes relevant metadata such as timestamps, event type, host information, and contextual details captured by the monitoring modules. This structured format ensures compatibility with backend analysis engines and supports efficient event processing, indexing, and correlation.

To enable external accessibility while maintaining security, the backend system can be exposed through a secure gateway such as a tunneling service. This approach allows the endpoint agent to communicate with the backend without directly exposing internal infrastructure, thereby reducing the attack surface. The communication design ensures that the backend remains protected while still allowing controlled access to incoming telemetry data.

The system also incorporates mechanisms to handle communication failures. In cases where the backend is temporarily unavailable, events are not discarded; instead, they are stored locally and transmitted once connectivity is restored, as described in the offline synchronization mechanism. This ensures that all collected telemetry is eventually delivered to the backend without data loss.

The integration with the backend enables advanced functionalities such as centralized monitoring, aggregation of events from multiple endpoints, and correlation of telemetry data to detect complex attack patterns. By forwarding both endpoint monitoring data and honeypot interaction events, the system supports comprehensive analysis and enhances situational awareness across the network.

The secure communication and integration design aligns with modern security architectures that emphasize encrypted data transmission, centralized analysis, and controlled exposure of services [21]. By combining secure telemetry transfer with reliable integration mechanisms, the endpoint agent ensures that collected data remains protected while enabling effective backend-driven analysis and response.

Overall, the secure communication and backend integration component ensures that the endpoint agent not only operates effectively at the host level but also contributes to a broader security ecosystem. This combination of local detection and centralized analysis enhances the overall effectiveness of the proposed system while maintaining strong security and reliability standards.

4.RESULTS AND DISCUSSIONS

4.1 Results (Demonstration Outcomes)

The proposed endpoint agent was evaluated through a series of controlled demonstration scenarios designed to validate its monitoring, detection, containment, and resilience capabilities. These demonstrations were conducted in a real-time execution environment to simulate practical attack and failure conditions.

During execution, the agent successfully demonstrated continuous monitoring of system activities across multiple domains. Process execution events were captured in real time along with associated metadata such as process identifiers, execution paths, and parent-child relationships. File system monitoring accurately detected file creation and modification events, while network monitoring provided visibility into active connections and their associated processes. These results confirm that the system is capable of generating comprehensive endpoint telemetry without interruption.

The deception-based SMB honeypot module effectively detected unauthorized access attempts. When the decoy resource was accessed, the system successfully logged the interaction and captured contextual information such as access time and system details. This demonstrates the effectiveness of embedding deception mechanisms directly within the endpoint to detect lateral movement and reconnaissance activities.

The signature-based detection mechanism was validated using controlled malware simulation. A test file with a known hash value was introduced into the system and executed. The agent successfully computed the SHA-256 hash, matched it against the IOC database, and triggered an automated containment action. The file was immediately moved to the quarantine directory, and the event was logged and recorded. This confirms the system's ability to perform real-time detection and local containment without reliance on external systems.

The offline operational capability of the system was tested by disabling network connectivity during execution. The agent continued to monitor system activities and store

events locally without interruption. Upon restoration of network connectivity, the system successfully synchronized the stored logs with the backend, demonstrating reliable offline logging and delayed transmission.

Resilience mechanisms were also validated through fault simulation. When the agent process was manually terminated, the watchdog mechanism successfully detected the failure and automatically restarted the agent, ensuring continuity of monitoring. Additionally, when the log file was deleted during execution, the system recreated the log file and resumed logging without data loss. These results confirm the robustness and survivability of the system under adverse conditions.

From a performance perspective, the system demonstrated efficient operation with minimal impact on system resources. Observations during testing indicate that CPU usage remained low during continuous monitoring, while memory consumption was stable and within an acceptable range for lightweight applications. This confirms that the system is suitable for deployment in resource-constrained environments such as SMEs.

Overall, the demonstration results validate that the proposed endpoint agent successfully achieves real-time monitoring, deception-based detection, local containment, offline operation, and resilient execution within a unified lightweight framework.

4.2 Research Findings

The evaluation of the proposed system provides several key findings related to the effectiveness and practicality of lightweight endpoint security solutions. One of the primary findings is that comprehensive endpoint visibility can be achieved without reliance on resource-intensive infrastructure. By combining process, file, network, and deception-based monitoring within a single agent, the system demonstrates that multi-layer telemetry collection is feasible in a lightweight implementation.

Another important finding is that embedding deception mechanisms directly within endpoint environments significantly enhances detection capability. The use of a low-interaction SMB honeypot allows the system to detect unauthorized access attempts and

lateral movement activities that may not be captured through conventional monitoring techniques. This highlights the effectiveness of integrating deception at the host level rather than relying solely on network-based honeypots.

The study also demonstrates that local detection and containment mechanisms can significantly reduce response time. By performing hash-based detection and quarantine directly at the endpoint, the system eliminates delays associated with centralized analysis and response. This is particularly important in mitigating fast-spreading threats such as malicious file execution or ransomware attacks.

Furthermore, the implementation confirms that resilience features such as offline logging and watchdog-based recovery are essential for maintaining continuous monitoring. The ability to operate independently during network disruptions and recover automatically from process failures ensures that security visibility is not compromised under real-world conditions.

In addition, the findings indicate that lightweight monitoring approaches can provide a practical balance between performance and functionality. Unlike traditional enterprise solutions, which may introduce significant overhead, the proposed system demonstrates that efficient monitoring and detection can be achieved with minimal resource consumption.

Overall, the research findings highlight that integrating monitoring, deception, detection, containment, and resilience within a single lightweight endpoint agent is a viable and effective approach for enhancing cybersecurity in SME environments.

4.3 Discussion

The results of this study demonstrate the effectiveness of the proposed endpoint agent in addressing key challenges associated with SME cybersecurity. By combining multiple security functionalities within a single lightweight system, the solution overcomes the limitations of fragmented and resource-intensive approaches commonly found in existing solutions.

One of the major strengths of the system is its ability to provide real-time monitoring and immediate response at the endpoint level. This reduces dependency on centralized infrastructures and improves detection-to-response time, which is critical in preventing damage from fast-moving threats. Additionally, the integration of a deception-based honeypot enhances the system's ability to detect sophisticated attacks that may bypass traditional monitoring mechanisms.

The implementation also demonstrates strong resilience and reliability through features such as offline operation, log continuity, and automated recovery. These capabilities ensure that the system remains functional and effective even under adverse conditions, making it suitable for real-world deployment.

However, certain limitations exist within the current implementation. The detection mechanism is primarily based on known signatures, which may not be effective against previously unseen or zero-day threats. While the system generates rich telemetry that can support advanced analysis, the current implementation does not include built-in machine learning-based anomaly detection. Additionally, the logging mechanism, while resilient, does not fully implement cryptographic tamper-proofing techniques, which could further enhance data integrity.

Despite these limitations, the proposed system provides a strong foundation for lightweight endpoint security. Future enhancements could include integration of anomaly detection models, advanced threat correlation techniques, and stronger cryptographic logging mechanisms to further improve detection accuracy and system robustness.

In conclusion, the discussion confirms that the proposed endpoint agent effectively addresses the identified research problem and demonstrates the feasibility of delivering comprehensive cybersecurity capabilities through a lightweight, endpoint-centric approach.

5. CONCLUSION

This report presented the design, implementation, and evaluation of a lightweight endpoint monitoring and containment agent developed as part of the ForLENS cybersecurity framework. The primary objective of this component was to provide continuous endpoint visibility, real-time detection, and immediate response capabilities within a resource-efficient architecture suitable for Small and Medium Enterprise (SME) environments.

The proposed system successfully integrates multiple security functionalities, including process monitoring, file system tracking, network activity observation, and deception-based detection through an embedded SMB honeypot. By combining these capabilities within a single endpoint agent, the system achieves comprehensive visibility into system behavior without relying on complex or resource-intensive infrastructure. This demonstrates that effective endpoint security can be implemented using a lightweight and modular approach.

A key contribution of this work is the implementation of local detection and containment mechanisms. The use of a signature-based detection approach, supported by an IOC database, enables the system to identify known threats and perform immediate quarantine actions directly at the endpoint. This significantly reduces response time and minimizes the potential impact of malicious activities. In addition, the integration of deception mechanisms enhances the system's ability to detect unauthorized access attempts and lateral movement at the host level.

The system also demonstrates strong resilience and reliability through features such as offline logging, automatic synchronization, and watchdog-based self-recovery. These mechanisms ensure continuous operation even under adverse conditions such as network disruptions or process failures. The ability to maintain monitoring and preserve log data under such conditions highlights the practicality of the proposed solution for real-world deployment.

Performance observations indicate that the agent operates efficiently with minimal impact on system resources, making it suitable for environments with limited computational capacity. This aligns with the goal of developing a cost-effective and easily deployable solution tailored for SMEs.

Although the current implementation focuses on signature-based detection and rule-based mechanisms, it provides a solid foundation for future enhancements. Potential improvements include the integration of machine learning-based anomaly detection, advanced correlation techniques, and stronger cryptographic logging mechanisms to further enhance detection accuracy and system security.

In conclusion, the proposed endpoint agent effectively addresses the challenges associated with lightweight endpoint security by providing an integrated solution that combines monitoring, deception, detection, containment, and resilience. The results demonstrate that a practical and scalable cybersecurity solution can be achieved at the endpoint level, making it a suitable approach for enhancing security in SME environments.

Reference List

- [1] S. Kandanaarachchi, H. Ochiai, and A. Rao, "Honeyboost: Boosting honeypot performance with data fusion and anomaly detection," arXiv preprint arXiv:2105.02526, 2021. [Online]. Available: <https://arxiv.org/abs/2105.02526>
- [2] J. Zou, Z. Sun, C. Ku, X. Li, and A. Dahbura, "WiP: Developing High-Interaction Honeypots to Capture and Analyze Region-Specific Bot Behaviors," in *Proc. Symp. Science of Security (HotSoS '24)*, Virtual, Apr. 2–4, 2024, pp. 1–9.
- [3] Y. L. Aung, Y. L. Khoo, D. Y. Zheng, B. S. Duo, S. Chattopadhyay, J. Zhou, L. Lu, and W. Goh, "HoneyWin: High Interaction Windows Honeypot in Enterprise Environment," *arXiv preprint arXiv:2505.00465v1*, May 2025, doi: 10.48550/arXiv.2505.00465.
- [4] A. Javadpour, F. Ja'fari, T. Taleb, M. Shojafar, and C. Benzaïd, "A comprehensive survey on cyber deception techniques to improve honeypot performance," *Computers & Security*, vol. 140, art. no. 103792, Mar. 2024, doi: 10.1016/j.cose.2024.103792.
- [5] M. Raz, P. V. S. Charan, P. Krishnamurthy, F. Khorrami, and R. Karri, "SHIELD: Secure Host Independent Extensible Logging for Tamper Proof Detection and Real Time Mitigation of Ransomware Threats," *arXiv preprint arXiv:2501.16619v2*, Apr. 2025, doi: 10.48550/arXiv.2501.16619.
- [6] K. Iyer, "Adaptive Honeypots: Dynamic Deception Tactics in Modern Cyber Defense," *International Journal of Science and Research Archive (IJSRA)*, vol. 4, no. 1, pp. 340–351, 2021, doi: 10.30574/ijrsra.2021.4.1.0210.
- [7] O. Kuznetsov, A. Rusnak, A. Yezhov, K. Kuznetsova, D. Kanonik, and O. Domin, "Evaluating the Security of Merkle Trees: An Analysis of Data Falsification Probabilities," *Cryptography*, vol. 8, no. 3, art. 33, Aug. 2024, doi: 10.3390/cryptography8030033.
- [8] D. Koisser and A. R. Sadeghi, "Accountability of Things: Large Scale Tamper Evident Logging for Smart Devices," *arXiv preprint arXiv:2308.05557*, Aug. 2023, doi: 10.48550/arXiv.2308.05557.
- [9] G. Guardiola Múzquiz and E. Soriano Salvador, "SealFSv2: Combining storage based and ratcheting for tamper evident logging," *International Journal of Information Security*, vol. 22, no. 2, pp. 447–466, 2023, doi: 10.1007/s10207-02200643-1.

- [10] H. Karlzén and T. Sommestad, "Automatic Incident Response Solutions: A Review of Proposed Solutions' Input and Output," in *Proc. 18th Int. Conf. Availability, Reliability and Security (ARES 2023)*, Benevento, Italy, Aug.–Sep. 2023, pp. 1–9, doi: 10.1145/3600160.3605066.
- [11] K. R. Ismail, Z. A. Brata, G. A. Nelistiani, S. Heo, H. Kim, and H. Kim, "Toward Robust Security Orchestration and Automated Response in Security Operations Centers with a Hyper-Automation Approach Using Agentic Artificial Intelligence," *Information*, vol. 16, no. 5, art. 365, 2025, doi: 10.3390/info16050365.
- [12] R. Holbel, J. Yerby, and W. Smith, "Utilizing Virtualized Honeypots for Threat Hunting, Malware Analysis, and Reporting," *Issues in Information Systems*, vol. 25, no. 1, pp. 265–278, 2024, doi: 10.48009/1_iis_2024_122.
- [13] S. Lanz, S. L.-R. Pignol, P. Schmitt, H. Wang, M. Papaioannou, G. Choudhary, and N. Dragoni, "Optimizing Internet of Things Honeypots with Machine Learning: A Review," *Applied Sciences*, vol. 15, no. 10, art. 5251, 2025, doi: 10.3390/app15105251.
- [14] F. Safarov, M. Basak, R. Nasimov, A. Abdusalomov, and Y. Im Cho, "Explainable Lightweight Block Attention Module Framework for Network-Based IoT Attack Detection," *Future Internet*, vol. 15, no. 9, art. 297, Sep. 2023, doi: 10.3390/fi15090297.
- [15] S. Roy, J. Li, B.-J. Choi, and Y. Bai, "A lightweight supervised intrusion detection mechanism for IoT networks," *Journal of Parallel and Distributed Computing*, vol. 166, pp. 112–123, 2022, doi: 10.1016/j.jpdc.2021.05.021.
- [16] M. Fatima, O. Rehman, I. M. H. Rahman, A. Ajmal, and S. J. Park, "Towards Ensemble Feature Selection for Lightweight Intrusion Detection in ResourceConstrained IoT Devices," *Future Internet*, vol. 16, no. 10, art. 368, Oct. 2024, doi: 10.3390/fi16100368.
- [17] V. Pham, E. Seo, and T.-M. Chung, "Lightweight Convolutional Neural Network Based Intrusion Detection System," *Journal of Communications*, vol. 15, no. 11, pp. 808–817, Nov. 2020, doi: 10.12720/jcm.15.11.808-817.
- [18] S. T. Muntaha, F. Ashraf, I. Shahzad, and J. Iqbal, "Designing an Adaptive Honeypot for Advanced Cybersecurity Threat Detection," *Spectrum of Engineering Sciences*, vol. 3, no. 5, pp. 816–847, May 2025.
- [19] Z. Morić, V. Dakić, and D. Regvart, "Advancing Cybersecurity with Honeypots and Deception Strategies," *Informatics*, vol. 12, no. 1, art. 14, Jan. 2025, doi:

10.3390/informatics12010014.

- [20] E. K. Isabirye, "Cloud Adoption and Digital Transformation Cybersecurity Consideration for SMEs," *Iconic Research and Engineering Journals*, vol. 8, no. 7, pp. 453, Jan. 2025.
- [21] H. Maosa, K. Ouazzane, and M. C. Ghanem, "A Hierarchical Security Event Correlation Model for Real-Time Threat Detection and Response," *Network*, vol. 4, no. 1, pp. 68–90, 2024, doi: 10.3390/network4010004.
- [22] Q. Wang, W. Ul Hassan, D. Li, K. Jee, X. Yu, K. Zou, J. Rhee, Z. Chen, W. Cheng, C. A. Gunter, and H. Chen, "You Are What You Do: Hunting Stealthy Malware via Data Provenance Analysis," in *Proc. 27th Network and Distributed System Security Symp. (NDSS)*, San Diego, CA, USA, Feb. 23–26, 2020, doi: 10.14722/ndss.2020.24167.
- [23] H. A. Balan and B. Agarwal, "Comparative Performance Evaluation of Modern Container Security Agents: Red Hat ACS, Wiz, SentinelOne, and Tenable," *Int. J. Computer Trends and Technology (IJCTT)*, vol. 73, no. 5, pp. 113–124, May 2025. [Online]. Available: <https://www.ijcttjournal.org/2025/Volume-73%20Issue-5/IJCTTV73I5P115.pdf>
- [24] M. AlKhanafseh and O. Surakhi, "Evidence Preservation in Digital Forensics: An Approach Using Blockchain and LSTM-Based Steganography," **Electronics**, vol. 13, no. 18, p. 3729, Sep. 2024, doi: 10.3390/electronics13183729.

Appendices

Appendix A : Scope of The Study

- The project focuses on SMEs with restricted financial resources and technical knowledge.
- The scope is restricted to endpoint-level protection utilizing surveillance and honeypot trickery.
- Only low-interaction honeypots are executed (not high-interaction, because of resource expenses).
- The research assesses resilience, stealth, and user-friendliness in simulated SME testbeds (e.g., VirtualBox/VMware).
- Integration with sophisticated AI/response frameworks (e.g., SOAR) is being evaluated, but implementation of full SOAR orchestration is beyond the range of this component.

Appendix B : Limitations

- Resource Limitation: Testing is restricted to a virtualized SME-style lab environment; real-world SME deployments may reveal additional performance challenges
- Prototype Scope: Focus is only on the lightweight endpoint agent with SMB honeypot functionality; integration with AI-driven analysis and dashboards belongs to other ForLens components.
- Attack Scope: Prototype validation is limited to common SME-relevant attacks (e.g., SMB exploitation attempts, unauthorized file access). Broader attack classes (e.g., nation-state APTs, large-scale DDoS) are outside scope.
- No Dataset Use: Evaluation relies on telemetry generated from honeypot engagements and simulated attacks in the lab; no external datasets are used.
- Time Constraint: The progression and assessment are limited to the duration of the project (approximately 8 months).

- Single Component Focus: Only the endpoint honeypot agent has been developed; more advanced AI modules are part of other team elements.

Appendix C : Risks and Mitgations

Technical Risks

- Risk: Detection of honeypots by sophisticated attackers.
Mitigation: Utilize covert operations, unpredictable services, and ongoing enhancements.
- Risk: Limited coverage of attack types beyond SMB-based deception. Mitigation: Focus prototype on SMB exploitation attempts, while leaving extension to other services for future work.
- Risk: Performance burden on SME equipment.
Mitigation: Aim for $\leq 10\%$ CPU and ≤ 150 MB RAM consumption.

Operational Risks

- Risk: SMEs might not have personnel with the expertise to set up solutions.
Mitigation: Provide plug-and-play deployment and simple configuration
- Risk: Intruders interfering with the agent.
Mitigation: Implement Self-recovery and tamper-resistant features.

Project Risks

- Risk: Limited real-world test data.
Mitigation: Utilize regulated ransomware/lateral movement exercises.
- Risk: Time constraints on development.
Mitigation: Break tasks into phases with clear deliverables (design, prototype, testing).

Appendix D : Ethical Considerations

- All evaluations are conducted in controlled lab environments using legal datasets (e.g., Metasploitable, Kali Linux).
- No genuine SME production systems or actual customer data will be utilized.
- Any collected telemetry will remain anonymized and securely stored.

- The research complies with IEEE Code of Ethics and institutional research policies.